

Hanoi University of Science and Technology
School of Information and Communications Technology



SOICT

DATA VISUALIZATION
PROJECT REPORT

Analyzing Global Tech Layoffs with Power BI

Nguyen Vu Trung Kien - 202416713

Data Science & AI, HUST

Tran Nam Hai - 202400103

Data Science & AI, HUST

Nguyen Cong Hung - 202416702

Data Science & AI, HUST

Nguyen Duc Hoai Anh - 202416662

Data Science & AI, HUST

Pham Tien Dung - 202416680

Data Science & AI, HUST

Course: Data Visualization

Class ID: 168068

Semester: 20252

Supervisor: Dr. Tran Viet Trung

Table of Contents

1	Introduction	4
2	Dataset Description	5
3	Data Preprocessing	6
4	Dashboard Design Overview	6
4.1	Page 1: Global Layoff Overview	6
4.2	Page 2: Company & Funding Stage Analysis	7
5	Technique Application by Chapter	9
5.1	Chapter 1: Overview of Data Visualization	9
5.1.1	Techniques / principles applied	9
5.1.2	How applied in the dashboard	9
5.2	Chapter 2: Visual Models and Encoding	9
5.2.1	Techniques / principles applied	9
5.2.2	How applied in the dashboard	9
5.2.3	Notes	10
5.3	Chapter 3: Graphical Perception	10
5.3.1	Techniques / principles applied	10
5.3.2	How applied in the dashboard	10
5.4	Chapter 4: Visualization for Multi-dimensional Data	10
5.4.1	Techniques / principles applied	10
5.4.2	How applied in the dashboard	10
5.4.3	Notes	11
5.5	Chapter 5: Visualization for Graphs	11
5.5.1	Notes	11
5.6	Chapter 6: Principles of Figure Design	11
5.6.1	Techniques / principles applied	11
5.6.2	How applied in the dashboard	11
5.7	Chapter 7: Map Visualization	11
5.7.1	Techniques / principles applied	11
5.7.2	How applied in the dashboard	12
5.8	Chapter 8: Interactive Visualization	12
5.8.1	Techniques / principles applied	12
5.8.2	How applied in the dashboard	12
5.9	Chapter 9: Storytelling with Data	12

5.9.1	Techniques / principles applied	12
5.9.2	How applied in the dashboard	13
5.10	Summary of Techniques	13
6	Economic Insight Analysis	14
6.1	Layoffs as a Signal of Macroeconomic Pressure	14
6.2	Industry-level Vulnerability	14
6.3	Geographic Concentration of Layoffs	14
6.4	Funding Stage and Layoff Severity	15
6.5	Funds Raised and Restructuring Pressure	15
6.6	Absolute vs. Relative Interpretation	15
7	Research Questions and Visualization Mapping	16
8	Limitations	16
9	Conclusion	17

1 Introduction

After a period of rapid expansion in the technology and startup ecosystem, many firms entered a challenging phase of global workforce reduction. This transition is closely associated with macroeconomic pressures, rising costs of capital, weaker market demand, and stronger requirements for operational efficiency. In this context, workforce reductions are no longer just isolated, internal company events. Instead, they act as recurring economic stress signals and critical indicators of broader global economic cycles, sector-level vulnerabilities, and the ultimate sustainability of growth models that depend heavily on external funding.

Despite the significance of these macroeconomic trends, layoff events are often reported as scattered headlines, leaving analysts without a unified, comprehensive view. To address this gap, this project utilizes the Kaggle Layoffs Dataset, which provides structured records of workforce reductions across global tech companies and startups. While the dataset captures essential metadata-such as company location, industry, funding stage, and the number of employees affected-these records remain fragmented observations without proper visualization. The primary goal of this project is to transform these scattered layoff records into structured, actionable economic insights.

To achieve this, we developed an analytical Power BI dashboard designed to consolidate these records into one clear view, revealing layoff patterns by time, industry, geography, and funding stage. The dashboard is structurally divided into two main analytical pages:

- **Page 1 - Global Layoff Overview:** provides a macro-level summary that answers critical questions about when the layoffs peaked, which industries were hit hardest, and where they were geographically concentrated.
- **Page 2 - Company & Funding Stage Analysis:** shifts to a micro-level perspective to explore the relationship between a company's maturity (funding stage), its accumulated capital, and the severity of its layoffs.

A core analytical requirement of this project is distinguishing between the absolute labor-market impact-measured by the total number of employees laid off-and the relative internal organizational severity, measured by the percentage of the workforce laid off. Rather than merely reporting aggregate numbers, the dashboard supports comparative and exploratory analysis for target audiences such as analysts, researchers, HR executives, and business decision-makers. Furthermore, this report highlights the data visualization rationale, visual encoding strategies, and economic insights derived from the dashboard, while also addressing analytical limitations such as public data bias, missing values, and the non-causal interpretation of funding stages.

2 Dataset Description

The project utilizes the Kaggle Layoffs Dataset, which tracks workforce reductions across the global tech ecosystem from early 2020 to the present. Each record represents a specific layoff event or announcement associated with a company. The primary variables utilized in the Power BI dashboard are summarized in Table 1.

Table 1: *Main variables used from the Kaggle Layoffs Dataset.*

Variable	Description
company	Name of the company that conducted a layoff.
location	City or local area associated with the company headquarter or layoff event.
industry	Business sector in which the company operates.
total_laid_off	Number of employees laid off in the event.
percentage_laid_off	Proportion of the company's workforce affected by the layoff.
date	Date on which the layoff event was recorded.
stage	Funding or development stage of the company.
country	Country associated with the company or layoff event.
funds_raised	Total amount of funding raised by the company, recorded in millions of USD.

Two variables are especially important for interpretation. The variable `total_laid_off` measures the absolute scale of a layoff event and therefore reflects the direct number of affected workers. In contrast, `percentage_laid_off` measures the relative severity with respect to the company's total size. A large, publicly traded technology company may lay off thousands of employees while only affecting a single-digit percentage of its workforce. Conversely, an early-stage startup may lay off fewer individuals, but that reduction could represent a substantial internal restructuring (e.g., 50% of its staff).

The variable `stage` allows the dashboard to compare companies across funding and maturity groups. It supports analysis of whether early-stage, late-stage, post-IPO, acquired, or other categories appear more exposed to restructuring pressure. Furthermore, The `funds_raised` variable provides critical context regarding a company's historical access to capital, previous growth expectations, and the scale of its expansion prior to restructuring.

3 Data Preprocessing

Before visualization, the raw dataset must be prepared to ensure that metrics and filters behave consistently within Power BI. The preprocessing and data modeling workflow follows a structured sequence:

- **Deduplication:** Remove duplicated rows to ensure identical layoff announcements do not artificially inflate aggregate numbers.
- **Standardization:** Standardize column names and enforce strictly correct data types. This includes converting the date column into a proper Date format and ensuring that `total_laid_off`, `percentage_laid_off`, and `funds_raised` are formatted as numeric values.
- **Categorical Validation:** Clean and validate text fields such as `industry`, `country`, and `stage` to ensure that conceptually identical categories are not fragmented by minor spelling or formatting inconsistencies.
- **Missing Value:** Address null values in `percentage_laid_off` and `funds_raised`. These are handled with caution, as blindly replacing them with zeros would severely distort the analytical reality of the dataset.

Preprocessing Interpretation

Because the dashboard intentionally juxtaposes absolute numbers (`total_laid_off`) with relative severity (`percentage_laid_off`), preprocessing must rigidly preserve the distinction between a missing value, a zero-layoff event, and a zero-percent reduction. Mishandling this distinction could lead to fundamentally misleading conclusions regarding company-level financial distress.

4 Dashboard Design Overview

With the dataset fully cleaned and structural missingness accounted for, the next step is to visually encode these records into an interactive format. The Power BI dashboard is organized into two analytical pages. Page 1 is designed for a macro-level overview of global trends, while Page 2 focuses on a micro-level perspective regarding company maturity and funding.

4.1 Page 1: Global Layoff Overview

Page 1 provides a high-level summary of global workforce reductions. It is designed to answer the core question: *When, in which industries, and in which geographic regions did layoffs occur most severely?*

The page incorporates the following visual components:

- **KPI Cards (Total Employees Laid off, Peak Monthly Layoff, Total Industries):** These

cards provide an immediate statistical snapshot before the viewer examines detailed charts. The *Peak Monthly Layoff* card specifically directs attention to the most critical period in the temporal trend.

- **Date Slicer:** Allows the viewer to dynamically filter layoff records by specific time periods.
- **Area Chart (Total Layoffs over Time):** Visualizes the temporal trend of layoffs, effectively highlighting periods of sudden spikes or declines over time.
- **Horizontal Bar Chart (Total Layoffs by Industry):** Compares the absolute scale of layoffs across different sectors, quickly identifying the hardest-hit industries.
- **Choropleth Map (Total Layoffs by Country):** Maps geographic distribution using color intensity (saturation). It utilizes tooltips to display additional context, such as the total number of companies affected in a specific region.

Together, these views support a compact yet multidimensional overview. The KPI cards establish the baseline scale, the area chart emphasizes temporal movement, the bar chart categorizes industry vulnerability, and the map expands the analysis into geographic space.

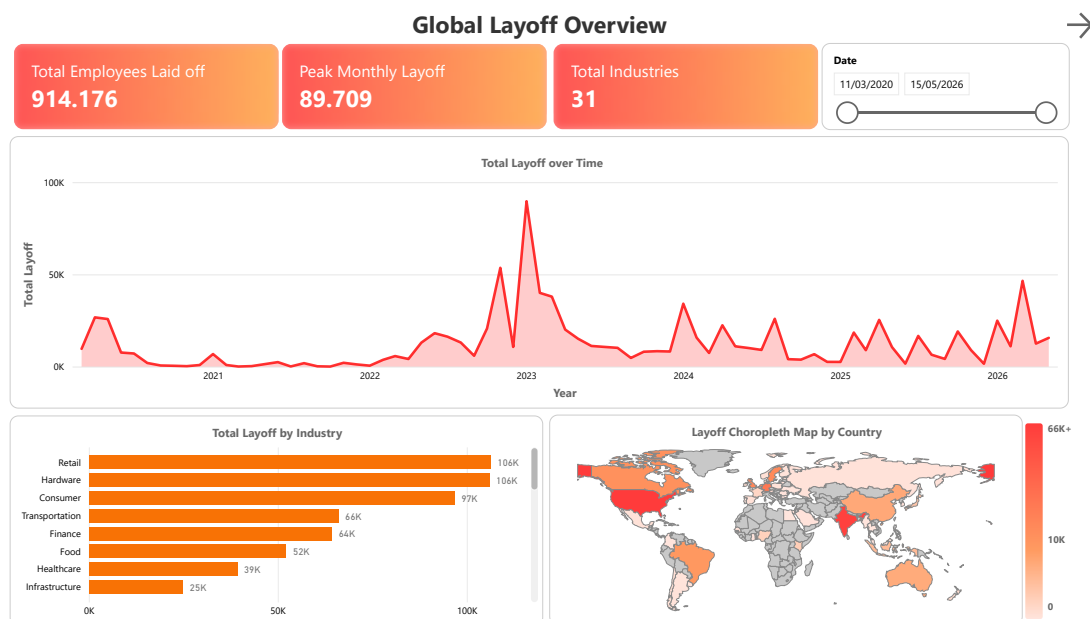


Figure 1: Overview dashboard summarizing global layoffs across time, industry, and geography. KPI cards provide aggregate measures, while the area chart, bar chart, and shape map support temporal, categorical, and spatial analysis.

4.2 Page 2: Company & Funding Stage Analysis

Page 2 shifts the focus to structural firm characteristics, examining the relationship between funding stage and layoff severity. It is designed to answer: *How does a company's funding stage relate to both the percentage and absolute scale of its layoffs?*

The page contains the following visual components:

- **KPI Cards (Total Employees Laid off, Total Funds Raised, Total Companies):** Summarize the scale, accumulated capital, and scope of companies under the selected filters.
- **Date Slicer:** Enables temporal filtering specific to the funding-stage analysis.
- **Vertical Bar Chart (Mean Layoff Percentage by Funding Stage):** Compares the relative layoff severity across different maturity stages, highlighting which stages faced the most intense internal restructuring.
- **Horizontal Bar Chart (Total Layoffs by Company):** Identifies individual companies contributing the largest absolute layoff counts, preventing the analysis from being skewed solely by aggregate stage averages.
- **Scatter Plot (Total Funding vs. Layoff by Stage):** Examines the correlation between accumulated funding and absolute layoff size. This reveals whether highly funded companies experienced proportionally smaller or larger layoffs.

The analytical strength of this page lies in its juxtaposition of relative and absolute metrics. One funding stage may exhibit a high `percentage_laid_off` (indicating severe internal disruption), while another may show a high `total_laid_off` (indicating broader labor-market impact). The inclusion of the company bar chart and scatter plot ensures that firm-level realities are not lost in high-level aggregations.

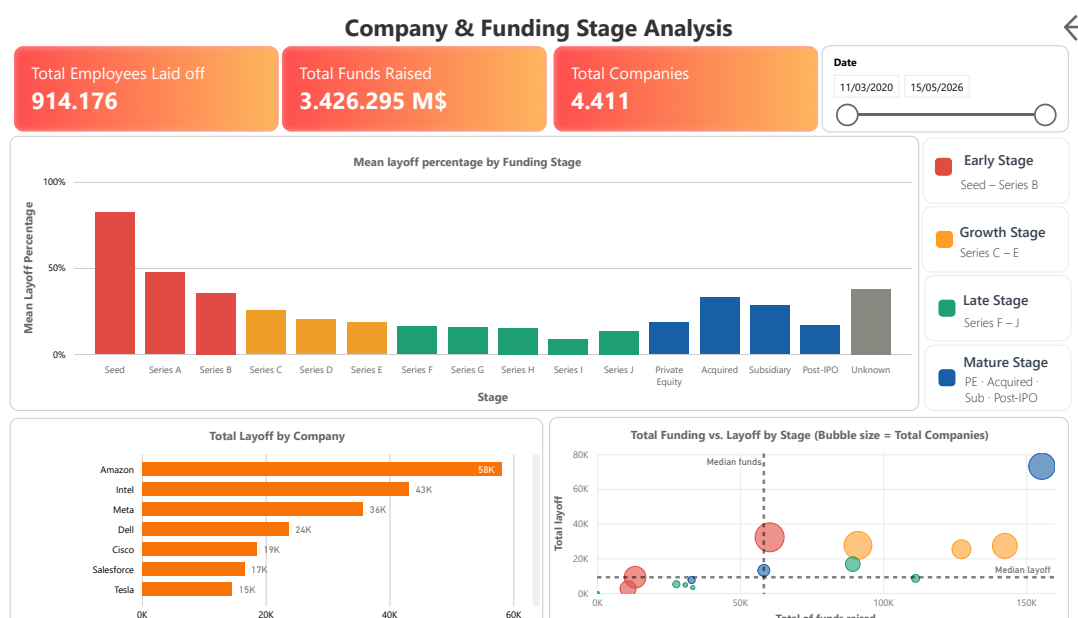


Figure 2: Funding-stage analysis dashboard examining layoff severity by company maturity stage. KPI cards summarize selected records, while the column chart, company bar chart, and scatter plot connect funding stage, firm-level scale, and the funding–layoff relationship.

5 Technique Application by Chapter

This section maps the visualization principles from the course chapter to the specific charts and layouts of the dashboard. Notably, Chapter 5 is addressed briefly to explain its omission.

5.1 Chapter 1: Overview of Data Visualization

5.1.1 Techniques / principles applied

- Why data visualization (Dataset selection rationale).
- Role of data visualization.
- Types of data visualization (Exploratory vs. Explanatory).

5.1.2 How applied in the dashboard

- **Dataset Rationale:** Layoffs are a timely economic phenomenon linking individual firm actions to broader labor-market cycles. In raw form, the dataset is a fragmented list; visualization is essential to aggregate these records and reveal macro-trends.
- **Explanatory Role:** The KPI cards and descriptive chart titles on both pages immediately communicate a clear summary (e.g., total affected employees, peak layoff month) to a reader scanning the dashboard.
- **Exploratory Role:** Slicers (Date) and Power BI's native cross-filtering allow the viewer to drill down and interrogate specific periods, sectors, and funding stages independently.

5.2 Chapter 2: Visual Models and Encoding

5.2.1 Techniques / principles applied

- Properties of data and measurement levels.
- Visual marks.
- Visual channels.

5.2.2 How applied in the dashboard

- **Data Types:** The fields company, industry, and country are nominal; stage is ordinal; total_laid_off, percentage_laid_off, and funds_raised are quantitative; date is temporal.
- **Marks & Channels:**
 - **Area Chart:** Uses line-and-area marks positioned on a temporal x-axis.
 - **Bar/Column Charts:** Use line marks (bars) where the channel of *length* encodes quantitative magnitude (total_laid_off).

- **Choropleth Map:** Uses region marks (country borders) with the channel of *color saturation/intensity* encoding magnitude.
- **Scatter Plot:** Uses point marks positioned on two quantitative Cartesian axes.

5.2.3 Notes

- The design deliberately favors the channels of *position* and *length* over *area* or *angle* (avoiding pie charts), aligning strictly with the standard ranking of visual channel effectiveness.

5.3 Chapter 3: Graphical Perception

5.3.1 Techniques / principles applied

- Magnitude estimation.
- Pre-attentive processing.
- Gestalt grouping principles.

5.3.2 How applied in the dashboard

- **Magnitude Estimation:** Bar charts were selected over pie charts because human perception judges linear length far more accurately than 2D angles.
- **Pre-attentive Processing:** The sharp visual peak in the Area Chart (late 2022 to early 2023) and the darkest colored countries on the Choropleth Map allow the viewer's eye to detect the most critical features instantly, before reading any specific data labels.
- **Gestalt Principles:** The KPI cards on both pages are arranged as a single horizontal strip. The principle of *Proximity* invites the viewer to read them as one cohesive summary unit, while *Similarity* in their visual styling reinforces this grouping.

5.4 Chapter 4: Visualization for Multi-dimensional Data

5.4.1 Techniques / principles applied

- Coordinate systems and axes.
- Visualizing amounts.
- Visualizing relationships.
- Visualizing trends.
- Visualizing uncertainty.

5.4.2 How applied in the dashboard

- **Amounts:** Horizontal Bar Charts (e.g., Total Layoffs by Industry/Company) accurately map absolute amounts using a Cartesian coordinate system.

- **Trends:** The Area Chart visualizes temporal trends, with the filled region emphasizing the accumulated volume of layoffs over time.
- **Relationships:** The Scatter Plot (Funds Raised vs. Total Layoffs) examines the correlation between capital backing and layoff severity, effectively revealing outliers.

5.4.3 Notes

- Uncertainty (e.g., missing values in `percentage_laid_off` due to unannounced private company data) is **not** visually encoded in the charts. This is a recognized limitation; instead of visualizing uncertainty, nulls were structurally excluded from averages to prevent skewing.

5.5 Chapter 5: Visualization for Graphs

5.5.1 Notes

- This chapter does not apply to this project. The Kaggle Layoffs dataset is purely tabular and contains no relational structure or networked entities (e.g., no interconnecting nodes/edges between companies). Therefore, layouts such as force-directed or hierarchical graphs have no meaningful analytical use here and were intentionally omitted.

5.6 Chapter 6: Principles of Figure Design

5.6.1 Techniques / principles applied

- The principle of proportional ink.
- Titles, captions, and tables.
- Problems with 3D charts.

5.6.2 How applied in the dashboard

- **Proportional Ink:** All bar and column charts strictly begin their value axes at zero. This ensures that the ink used (length of the bar) remains perfectly proportional to the data value.
- **No 3D Charts:** 3D effects were completely avoided to prevent spatial distortion.
- **Titles & Consistency:** Every chart features a highly descriptive title and appropriately labeled axes. A consistent red/orange color palette runs through the dashboard, creating a cohesive visual identity suited to the “warning” nature of layoffs.

5.7 Chapter 7: Map Visualization

5.7.1 Techniques / principles applied

- Types of map data visualization.

5.7.2 How applied in the dashboard

- **Map Type:** A Choropleth Map (Shape map in Power BI) is utilized on Page 1 to show geographic distribution.
- **Encoding:** The data (`total_laid_off`) is encoded sequentially onto the map regions through color intensity (darker red implies more layoffs). Tooltips are added to provide context (Total Companies affected) without overcrowding the map.

5.8 Chapter 8: Interactive Visualization

5.8.1 Techniques / principles applied

- Interaction techniques: filtering, selection, view transformation.
- Tools and libraries.

5.8.2 How applied in the dashboard

- **Filtering & Selection:** Date slicers are implemented for temporal filtering, while visual elements like the Choropleth map actively function as spatial filters. Utilizing Power BI's native cross-filtering capability ensures that selecting a specific industry, funding stage, or country dynamically updates all other charts and KPI cards to reflect that specific segment.
- **View Transformation:** Navigation buttons are provided to seamlessly transform the view between the macro overview (Page 1) and the micro funding-stage analysis (Page 2), allowing users to switch analytical contexts effortlessly.
- **Tool:** Microsoft Power BI was utilized as the primary BI tool to construct and deploy these interactive features.

Notes

- Animation was not utilized. Because layoff records are discrete, point-in-time events rather than continuous physics or motion data, animating them would add cognitive load without adding analytical value.

5.9 Chapter 9: Storytelling with Data

5.9.1 Techniques / principles applied

- Fundamental principles of data storytelling.
- Narrative styles in data storytelling.
- Interaction and exploration in storytelling.

5.9.2 How applied in the dashboard

- **Story Structure:** The dashboard forms a clear narrative arc from Macro to Micro. Page 1 sets the scene (the “What, When, and Where”), while Page 2 deepens the account by examining structural causes (the “Why and How” related to funding stages and company maturity).
- **Narrative Style:** The dashboard uses a “Martini Glass” narrative structure. It begins with an authored, guided sequence (the macro overview and KPI highlights) but opens up to allow free exploration.
- **Interaction in Storytelling:** Through the slicers and cross-filtering, the reader can follow the intended story or branch off to test their own hypotheses, successfully balancing an authored narrative with reader-driven analysis.

5.10 Summary of Techniques

Table 2 consolidates the techniques discussed above, the dataset variables each one encodes, and the analytical role it plays in the dashboard.

Table 2: *Summary of visualization techniques used in the Power BI dashboard.*

Technique	Main Variables	Analytical Role
KPI cards	total_laid_off, Peak Month, industry, company, funds_raised	Summarize overall scale, peak timing, coverage, and funding context before detailed chart reading.
Date slicer	date	Enables interactive filtering and comparison of different economic periods.
Area chart	date, total_laid_off	Reveals temporal trends, peaks, and periods of intensified layoffs.
Clustered bar chart	industry, company, total_laid_off	Compares absolute layoff scale across categories and identifies top contributors.
Shape map	country, total_laid_off, Total Company	Shows geographic concentration and adds company-count context through tooltip interaction.
Column chart	stage, percentage_laid_off	Compares relative layoff severity across company maturity and funding stages.

Technique	Main Variables	Analytical Role
Scatter plot	funds_raised, total_laid_off	Examines the relationship between funding and layoff scale and highlights outliers.

6 Economic Insight Analysis

Building upon the visual and narrative techniques discussed in the previous section, the dashboard enables deep economic interpretation by connecting layoff scale, timing, industry, geography, funding stage, and company-level structure. The insights below synthesize the macro and micro-level realities extracted from the Power BI dashboard.

6.1 Layoffs as a Signal of Macroeconomic Pressure

When total layoffs increase sharply over time, it reflects pressure from the macroeconomic environment—such as higher interest rates, increased cost of capital, weaker consumer demand, and heavily revised growth expectations. Particularly, technology companies depend on expectations of future growth. When capital markets tighten, these firms often reduce operating costs more aggressively than firms in stable, traditional sectors.

Peaks in the area chart highlight these periods of synchronized restructuring. Notably, the dashboard reveals a massive spike during **late 2022 to early 2023**, suggesting that the global tech ecosystem collectively adjusted its workforce size in response to an abrupt end to the pandemic-era zero-interest-rate environment.

6.2 Industry-level Vulnerability

The industry bar chart demonstrates that layoffs are not evenly distributed across sectors. Industries with the highest total layoffs are typically those that expanded aggressively during the preceding growth period and faced severe corrections when business models could no longer sustain those expansion rates.

The dashboard indicates that **Retail** (closely followed by Consumer and Hardware) contributes the largest absolute share of layoffs. This suggests that e-commerce and consumer-facing tech were heavily exposed to post-pandemic corrections and cost restructuring. This sectoral concentration proves that workforce reductions were structurally driven rather than randomly distributed.

6.3 Geographic Concentration of Layoffs

The choropleth map illustrates the geographic footprint of the layoffs. However, a country with high total layoffs could represent a massive tech ecosystem with thousands of firms, or

merely a few dominant companies executing massive cuts.

The tooltip detailing *Total Companies* contextualizes this. The **United States** emerges as the most intensely shaded region, accompanied by a dominant company count. This combination indicates systemic exposure across the entire US tech sector, rather than isolated events by a few megacorporations, cementing its position as the epicenter of the layoff wave.

6.4 Funding Stage and Layoff Severity

Page 2 confirms that a company's funding stage heavily influences its layoff severity profile. Early-stage companies (e.g., Seed, Series A) exhibit higher *layoff percentages* because they operate with limited cash reserves, fragile business models, and face existential threats when unable to raise new venture capital. Conversely, Late-stage or Post-IPO companies exhibit larger *absolute layoff counts* because they employ tens of thousands of people and face relentless shareholder pressure to optimize profitability.

Therefore, funding stage is not merely a descriptive label; it is a proxy for company maturity, capital access, and structural resilience.

6.5 Funds Raised and Restructuring Pressure

The KPI for Total Funds Raised and the scatter plot contextualize layoffs against historical investment. High accumulated funding does not immunize a firm from layoffs; paradoxically, it often indicates that the firm previously scaled aggressively and later required severe right-sizing when growth expectations normalized.

With total funding reaching approximately **\$3.43 Trillion USD** across **4,411** companies, the scatter plot highlights critical outliers: several massively funded “unicorn” companies still executed large-scale layoffs. This proves that for many firms, layoffs were a strategic correction of over-hiring rather than a pure symptom of bankruptcy.

6.6 Absolute vs. Relative Interpretation

Ultimately, the dashboard's analytical strength lies in distinguishing between *Total Layoffs* and *Percentage Laid Off*. Total layoffs measure the sheer volume of affected workers, capturing the external macroeconomic shock to the labor market. Percentage laid off measures the internal severity of the restructuring.

A multinational tech giant may record a staggering total layoff count while only reducing its workforce by 5%. In contrast, a small startup might lay off just 50 people, but if that represents 80% of its staff, it signals deep distress or imminent closure. This duality is central to interpreting the dashboard accurately.

7 Research Questions and Visualization Mapping

Table 3 explicitly maps the core research questions to the dataset variables, chosen visualization techniques, visual encodings, and their specific analytical purposes.

Table 3: Mapping between research questions, dataset variables, visual encodings, and analytical purposes.

Research Question	Dataset Variables	Visualization Technique	Visual Encoding	Analytical Purpose
How has the total number of layoffs changed over time?	date, total_laid_off	Area chart	x-axis = date; y-axis = total_laid_off	Identify temporal trends and macro peaks.
Which industries experienced the largest absolute layoffs?	industry, total_laid_off	Clustered bar chart	category = industry; bar length = total_laid_off	Compare sector-level impact.
Which countries are most affected by workforce reductions?	country, total_laid_off, Total Company	Choropleth map with tooltips	region = country; color intensity = total_laid_off; tooltip = Total Company	Analyze spatial concentration and scope.
How does funding stage relate to relative layoff severity?	stage, percentage_laid_off	Column chart	x-axis = stage; y-axis = mean percentage_laid_off	Evaluate internal distress across maturity stages.
Which specific companies contribute the most to the layoff volume?	company, total_laid_off	Clustered bar chart	category = company; bar length = total_laid_off	Identify dominant firm-level contributors.
Does high historical funding protect against large-scale layoffs?	funds_raised, total_laid_off	Scatter plot	x = funds_raised; y = total_laid_off	Examine the capital-layoff correlation and outliers.

8 Limitations

While the dashboard provides a highly effective exploratory view of global technology layoffs, viewers must interpret the visualizations with an understanding of the following analytical and data-related limitations:

- **Data Collection Bias:** The Kaggle Layoffs dataset relies heavily on media reports, press releases, and public disclosures. This inherently skews the data toward publicly traded technology giants and high-profile startups, while smaller private firms often conduct workforce reductions without public announcement.
- **The Disproportionate Weight of Megacorporations:** Absolute metrics, specifically `total_laid_off`, are fundamentally constrained by the baseline size of the company. A few multinational tech giants can easily dominate aggregate layoff counts, the industry bar charts, and the geographic choropleth map. This heavy skew can mask the severe internal distress of smaller startups if viewers only look at absolute volume. Therefore, total layoffs must always be consciously juxtaposed with the percentage laid off to separate broad market impact from

internal organizational severity.

- **Exploratory Scope and Non-Causal Associations:** The Power BI dashboard is designed strictly for exploratory data analysis (EDA) and pattern recognition. While it effectively highlights correlations—such as identifying which funding stages exhibit the highest relative layoff severity—it cannot establish statistical causality. Variables like `stage` or `funds_raised` serve as descriptive grouping dimensions, not proven triggers. The visualizations help generate hypotheses regarding macroeconomic pressures, but they do not definitively prove the causal mechanisms behind specific corporate restructuring events.

9 Conclusion

This report has detailed the design, rationale, and implementation of an analytical Power BI dashboard tailored to investigate the global wave of layoffs. By rigorously applying foundational data visualization principles, the project successfully transformed a fragmented, tabular dataset into a cohesive analytical narrative.

The dual-page architecture systematically guides the viewer from macro to micro perspectives. While Page 1 establishes the global context, identifying the precise temporal peaks of the layoff wave, mapping its geographic epicenter, and highlighting the disproportionate vulnerability of specific sectors like Retail. Page 2 deepens this analysis by probing the structural characteristics of the firms themselves, revealing how a company's funding maturity and accumulated capital dictate its restructuring behavior.

Ultimately, this project demonstrates that when raw data is strictly prepared, modeled, and visually encoded with precision, it transitions from a simple historical record into a powerful diagnostic instrument. It empowers analysts, researchers, and strategic decision-makers to navigate complex economic signals, evaluate firm-level vulnerabilities, and extract actionable insights from global labor-market fluctuations.